Backbone Detectives · Staff Answers

Learning goal: I can use inside body clues to sort tricky animals and explain my choice.

Arrival Answers

1. Yes. A snake has a long backbone.
2. No. A worm has a soft body.
3. A hard case on the outside.
4. Yes. You can ask or pass.

Starter & Hinge Answers

Starter: A. Snakes and fish have no legs, yet they have a backbone inside.

Hinge: A. A snake is a vertebrate because it has a backbone inside.

Evidence Desk Cards

Snake x-ray: Vertebrate: the x-ray shows a spine inside.

Bat x-ray: Vertebrate: a bat has a spine inside.

Lobster clue: Invertebrate: hard case outside and no spine.

Jellyfish clue: Invertebrate: soft body with no bones.

Beetle clue: Invertebrate: its hard case is on the outside.

Closed box photo: Ask for inside clue: we cannot see the body yet.

Check & Repair Answer

Wings do not prove a backbone. A bird has a spine inside, while a butterfly has no bones.

Exit Ticket Answers

1. An internal backbone (spine). 2. Snake = vertebrate (spine inside); crab = invertebrate (hard shell outside, no spine). 3. Legs, wings, or size do not show whether there is a spine inside.